USBL POSITIONING AND COMMUNICATION SYSTEM

S2CR 18/34H USBL

PRODUCT INFORMATION

Simultaneous positioning and communication

S2C Technology: accurate 3D positioning and reliable data transmissions with up to 13.9 kbit/s

Hemispherical beam pattern, optimized for medium range transmissions in vertical and slant channels

TECHNICAL SPECIFICATIONS

GENERAL	OPERATING DEPTH	Delrin	200 m
		Aluminium Alloy	1000 m
		Stainless Steel	2000 m
		Titanium	2000 m
	OPERATING RANGE		3500 m
	FREQUENCY BAND		18 - 34 kHz
	TRANSDUCER BEAM PATTERN		hemispherical
USBL	SLANT RANGE ACCURACY [1)]		0.01 m
	BEARING RESOLUTION		0.1 degrees
	NOMINAL SNR		10 dB
CONNECTION	ACOUSTIC CONNECTION		up to 13.9 kbit/s
	BIT ERROR RATE		less than 10^{-10}
	INTERNAL DATA BUFFER		1 MB, configurable
	HOST INTERFACE [2)]		Ethernet, RS-232 (RS-485/422*)
	INTERFACE CONNECTOR		up to 2 SubConn® Metal Shell 1500 Series
POWER	CONSUMPTION	Stand-by Mode	2.5 mW
		Listen Mode [3)]	5 - 285 mW
		Receive Mode [4)]	1.4 W
		Transmit Mode	2.8 W, 1000 m range
			8 W, 2000 m range
			35 W, 3500 m range
			65 W, max. available
	POWER SUPPLY [5)]		External 24 VDC (12 VDC*) or internal rechargeable battery*
PHYSICAL	DIMENSIONS [6)]	Housing/USBL antenna	∅ 113 mm x 218 mm / ∅ 180 mm x 165 mm
		Total length	383 mm
	WEIGHT dry/wet	Delrin	5800*/1000* g
		Aluminium Alloy	6000*/1800* g
		Stainless Steel	14000/8000 g
		Titanium	11000*/5800* g

* optional
1) Slant range estimation is based on the measured time delay, slant range accuracy depends on sound velocity profile, refraction and signal-to-noise ratio.
2) See the Configuration Options for available standard interface combinations.
3) User-configurable Listen Mode is only available with a Wake-Up module installed. Power consumption in Listen Mode depends on Listen Mode settings.
4) Power consumption for the RS-232 interface option. Add 500 mW for the Ethernet interface option. Add 300 mW for Wake-Up Module.
5) Contact EvoLogics for more information on power supply options.
6) Dimensions of a Delrin housing, other builds are slightly larger. Marked* weights are estimates.

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PRODUCT INFORMATION

APPLICATIONS

Positioning, navigation and communication for AUVs and ROVs

Underwater acoustic sensor networks

CONFIGURATION OPTIONS

HOUSING	DELRIN	Plastic non-magnetic corrosion-resistant housing for short-term deployments, depth rating 200 m	
HOUSING	ALUMINIUM ALLOY	Light metal housing for short-term deployments, depth rating 1000 m	
HOUSING	STAINLESS STEEL	Robust metal, suitable for long-term deployments in harsh environments, depth rating 2000 m	
HOUSING	TITANIUM	Corrosion resistant, suitable for long-term deployments in harsh environments, depth rating 6000 m	
INTERFACE	1 CONNECTOR	RS-232 [1] or	
INTERFACE		Ethernet	
INTERFACE	2 CONNECTORS	RS-232 + RS-232 or	
INTERFACE		RS-232 + Ethernet	
MODULES	WAKE-UP MODULE [2]	RS-232 interface	✓
MODULES		Ethernet interface	×
MODULES		RS-232 + RS-232 interface	✓
MODULES		RS-232 + Ethernet interface	×
MODULES	ROLL, PITCH, HEADING [3]	internal AHRS, Xsens® MTx	

[1] One RS-232 Interface can be replaced with either RS-485 or RS-422 interface. More interface configurations available by special request. Contact EvoLogics for more information.
[2] The Wake Up Module turns the rest of the device on if it detects incoming acoustic signals or incoming data on the host interface. Once the device completes receiving or transmitting data, it switches itself off.
[3] Power consumption increases by 800 mW with an AHRS installed.

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